



Red Hat AMQ Broker 7.14

Release Notes for Red Hat AMQ Broker 7.14

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Abstract

These release notes contain the latest information about new features, enhancements, fixes, and issues contained in the AMQ Broker 7.14 release.

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MAKING OPEN SOURCE MORE INCLUSIVE

Red Hat is committed to replacing problematic language in our code, documentation, and web properties. We are beginning with these four terms: master, slave, blacklist, and whitelist. Because of the enormity of this endeavor, these changes will be implemented gradually over several upcoming releases. For more details, see [our CTO Chris Wright's message](#).

CHAPTER 1. LONG TERM SUPPORT FOR AMQ BROKER 7.14

AMQ Broker 7.14 has been designated as a Long Term Support (LTS) release version. For details of the terms of an LTS release, see [How long are AMQ LTS releases supported?](#)

Support for Red Hat Enterprise Linux and OpenShift Container Platform

The AMQ Broker 7.14 LTS version supports:

- Red Hat Enterprise Linux 8, 9 and 10
- All OpenShift Container Platform releases currently in the full or maintenance support phases, as well as releases from version 4.14 onward that offer extended support.
- Microsoft Windows Server 2019, 2022 and 2025

Red Hat strives to ensure that AMQ Broker remains compatible with future versions of OpenShift Container Platform; however this compatibility cannot be guaranteed. Interoperability testing is performed for each new OpenShift Container Platform version. If no compatibility issues are found, the new OpenShift Container Platform version is added to the [Red Hat AMQ Broker 7 Supported Configurations](#).

CHAPTER 2. SUPPORTED CONFIGURATIONS

For information on supported configurations, see [Red Hat AMQ Broker 7 Supported Configurations](#) .

Java versions supported

AMQ Broker 7.14 supports Java 17, 21 and 25.

Operator compatibility with broker versions

An AMQ Broker 7.14 Operator can manage any of the following broker versions: 7.12.x, 7.13.x and 7.14.x.

Openwire support

AMQ 7 Broker has provided support for the Openwire protocol since its release in 2017 as a means to migrate client applications to AMQ 7. With the release of AMQ Broker 7.9.0 in 2021, the Openwire protocol was deprecated and customers were encouraged to migrate their existing Openwire client applications to one of the fully supported protocols of AMQ 7 (CORE, AMQP, MQTT, or STOMP). Starting with the AMQ Broker 8.0 release, the Openwire protocol will be removed from AMQ Broker.

CHAPTER 3. NEW AND CHANGED FEATURES

This section describes a highlighted set of enhancements and changed features in AMQ Broker 7.14. For a full list of enhancements, see [AMQ Broker 7.14.0 Enhancements](#)

Option to manage scaledown by using the AMQ Broker Operator

You can configure the AMQ Broker Operator to manage a cluster scaledown. When the Operator manages a cluster scaledown, a target broker is automatically chosen for messages that must be migrated. The pod which is being removed remains in the StatefulSet until all messages are moved to a remaining broker in the cluster. For more information, see [Enabling message migration to support cluster scaledown](#) in *Deploying AMQ Broker on OpenShift*.

Security settings can be applied to temporary queues

The default `<security-setting match="#">` applies any configured security settings to *all* queues, including temporary ones. In 7.14, you can configure specific security settings for temporary queues only. For more information, see [Applying specific security settings to temporary queues](#) in *Configuring AMQ Broker*.

OpenID Connect (OIDC) support in Hawtio for integrating AMQ Broker with Red Hat Single Sign On

To handle user authentication in AMQ Management Console, you can use the built-in OpenID Connect (OIDC) support in Hawtio to integrate AMQ Broker with Red Hat Single Sign-On. For more information, see [Securing AMQ Management Console using Red Hat Single Sign-On](#) in *Managing AMQ Broker*.

Default security manager and default domain configurable in broker custom resource (CR)

In the broker CR, you can set Java system properties to change the default security manager and the default security domain that is used for user authentication. In previous releases, an init container was required for this configuration. For more information, see [Changing the default security manager](#) and [Changing the default JAAS security domain](#) in *Deploying AMQ Broker on OpenShift*.

Option to delete paged folders after use

The first time the broker pages messages for an address, it creates and retains a paging directory for that address. A large number of paging directories can delay the broker startup. To reduce startup time, you can set a **purge-page-folders** parameter to delete the paging directory for each address after use. For more information, see [Deleting paging directories for individual addresses](#) in *Configuring AMQ Broker*.

Persistent volumes are mounted in init containers

If you use the **extraVolumes** and **extraVolumeMounts** attributes to add a persistent volume and mount the volume in each broker container, the volumes are also mounted in any init container that is configured. You can then, for example, set **readOnlyRootFilesystem: true** for the init container.

Variables supported in a `spec.resourceTemplate.patch` attribute

You can use the following variables for string values in the **spec.resourceTemplate.patch** attribute: (CR_NAME), (CR_NAMESPACE), (BROKER_ORDINAL), (ITEM_NAME), (RES_TYPE) and (INGRESS_DOMAIN). For an example, see [Configuring role-based access control for management operations](#) in *Deploying AMQ Broker on OpenShift*.

Parameters added to customize the Online Certificate Status Protocol (OCSP) revocation checking

You can use the following parameters to customize OCSP revocation checking: **crlPath**, **crcOptions** and **ocspResponderURL**. For more information, see [Acceptor and Connector Configuration Parameters](#) in *Configuring AMQ Broker*.

Parameter added to disable http/2 in client connections to the embedded web server that hosts AMQ Management Console

You can set the **http2** parameter to "false" in the **bindings** element that defines how clients connect to the web server that hosts AMQ Management Console.

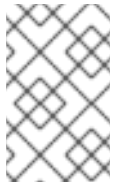
Support for encoding the URL of a JDBC database

If you configure the broker to persist message data in a JDBC database, you can now encrypt the URL of the database in the broker configuration file. For more information, see [Encrypting a JDBC URL in a configuration file](#) in *Configuring AMQ Broker*.

Operator channels

A new Operator, **Red Hat Integration - AMQ Broker for RHEL 9 (Multiarch)**, is provided on OperatorHub for 7.14. Updates for this Operator are provided in the **7.14.x** channel only. The **7.14.x** channel is a Long Term Support (LTS) channel.

A **7.13.x** LTS channel provides updates for the **Red Hat Integration - AMQ Broker for RHEL 9 (Multiarch) Operator**. This channel is for version 7.13 only.



NOTE

It is not possible to upgrade the Operator by switching channels. You must uninstall the existing Operator and install the new version of the Operator from the appropriate channel.

To determine which Operator to choose, see the [Red Hat Enterprise Linux Container Compatibility Matrix](#).

Maven groupId

Starting in 7.14, the Maven **groupId** for AMQ Broker modules changed from **org.apache.activemq** to **org.apache.artemis**.

Example programs

In AMQ Broker 7.14, example programs are not distributed and installed with the broker. Instead, you can access the example programs in the following repository: <https://github.com/apache/artemis-examples>.

CHAPTER 4. DEPRECATED FEATURES

This section describes features that are supported, but have been deprecated from AMQ Broker.

Scaledown by using the scaledown controller

Starting in 7.14, the default method of scaling down the number of broker in cluster by using the scaledown controller is deprecated. You can now use the AMQ Broker Operator to manage cluster scaledown. For more information, see [Managing scaledown by using the Operator](#) in *Deploying AMQ Broker on OpenShift*.

Keycloak login modules

Starting in 7.14, the Keycloak login modules included in AMQ Broker are deprecated. For applications that support OAuth authentication, such as the AMQ Management Console, you can use OpenID authentication instead of using the Keycloak BearerTokenLoginModule for user authentication. For more information, see [Securing AMQ Management Console using Red Hat Single Sign-On](#) in *Managing AMQ Broker*.

Two-way algorithm of the default codec

Starting in 7.13, the two-way algorithm of the default codec, **DefaultSensitiveStringCodec**, is deprecated.

ActiveMQArtemisAddress CRD

Starting in 7.12, the **ActiveMQArtemisAddress** CRD is deprecated. Use the **spec.brokerProperties** attribute in the **ActiveMQArtemis** CR to create addresses and queues for your deployment.

ActiveMQArtemisSecurity CRD

Starting in 7.12, the **ActiveMQArtemisSecurity** CRD is deprecated. Use the **spec.brokerProperties** attribute in the **ActiveMQArtemis** CR to configure security for your deployment.

ActiveMQArtemisScaledown CRD

Starting in 7.12, the **ActiveMQArtemisScaledown** CRD is deprecated. The **ActiveMQArtemisScaledown** CRD is used internally by the broker, so this change is transparent to AMQ Broker administrators.

Connection pooling for LDAP queries

Starting in 7.12, the **connectionPool** parameter, which enables connection pooling for LDAP queries, is deprecated. The built-in authorization and authentication caches provide an alternative way to optimize the performance of LDAP queries. For information on customizing the built-in caches, see [Configuring authentication and authorization caching](#).

upgrade attribute in Custom Resource

Starting in 7.11, the **upgrade** attribute and the associated **enabled** and **minor** attributes are deprecated because they cannot work as originally designed. Use the **image** or **version** attributes to deploy specific broker container images.

queues configuration element

Starting in 7.10, the <queues> configuration element is deprecated. You can use the <addresses> configuration element to create addresses and associated queues. The <queues> configuration element will be removed in a future release.

getAddressesSettings method

Starting in 7.10, the **getAddressesSettings** method, which is included in the `org.apache.activemq.artemis.core.config.Configuration` interface, is deprecated. Use the **getAddressSettings** method to configure addresses and queues for the broker programmatically.

OpenWire protocol

Starting in 7.9, the OpenWire protocol is a deprecated feature. If you are creating a new AMQ Broker-based system, use one of the other supported protocols. Starting with the 8.0 release, the Openwire protocol will be removed from AMQ Broker.

Adding users when broker instance is not running

Starting in 7.8, when an AMQ Broker instance is not running, the ability to add users to the broker from the CLI interface is removed.

async-connection-execution-enabled parameter

Starting in 7.5, the **async-connection-execution-enabled** parameter is deprecated. This parameter allows you to disable asynchronous connection execution.

Network pinger

Starting in 7.5, network pinging is a deprecated feature. Network pinging cannot protect a broker cluster from network isolation issues that can lead to unrecoverable message loss. This feature will be removed in a future release. Red Hat continues to support existing AMQ Broker deployments that use network pinging. However, Red Hat no longer recommends use of network pinging in new deployments. For guidance on configuring a broker cluster for high availability and to avoid network isolation issues, see [Implementing high availability](#) in *Configuring AMQ Broker*.

Hawtio dispatch console plugin

Starting in 7.3, AMQ Broker no longer includes the Hawtio dispatch console plugin, **dispatch-hawtio-console.war**. Previously, the dispatch console was used to manage AMQ Interconnect. However, AMQ Interconnect now uses its own, standalone web console.

CHAPTER 5. TECHNOLOGY PREVIEW

This section describes Technology Preview features in AMQ Broker 7.14, and other Red Hat Integration products.



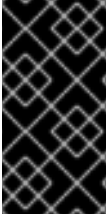
IMPORTANT

Technology Preview features are not supported with Red Hat production service-level agreements (SLAs) and might not be functionally complete. Red Hat does not recommend using them for production. For more information, see [Red Hat Technology Preview Features Support Scope](#).

Lock coordinator to coordinate starting acceptors among brokers

The Lock Coordinator provides a distributed locking mechanism to coordinate starting acceptors among broker instances. If an acceptor is configured to use a lock coordinator, the broker must acquire the distributed lock before the acceptor starts. In a mirror configuration, this feature ensures that clients connect to only one broker at a time, which prevents split-brain and duplicate message processing. For information on how to use this feature, see the [Apache Artemis documentation](#).

CHAPTER 6. DEVELOPMENT PREVIEW



IMPORTANT

Developer Preview versions are provided to expose features from upstream communities for use by early adopters and integrators to explore and interact with new capabilities in advance of their possible inclusion in a Red Hat product offering. For more information, see [Developer Preview Support Scope](#).

Restricted mode

In 7.14, the **ActiveMQArtemis** CR supports a new attribute, **spec.Restricted**. If this attribute is set to **True**, an empty embedded broker is deployed. You can configure this broker by using the **brokerProperties** attribute in the **ActiveMQArtemis** CR. To secure end-to-end communication the broker requires a public key infrastructure (PKI). You can use cert-manager to generate the TLS certificates for the PKI.

Self-provisioning plugin

The self-provisioning plugin is a dynamic plugin for the OpenShift Console. The plugin adds a new broker workload type to the OpenShift console that allows you to create and manage broker deployments on OpenShift. The self-provisioning plugin is available on the [Software Downloads page for AMQ Broker 7.14 releases](#):



NOTE

You can install the self-provisioning plugin only on a cluster that is running OpenShift 4.17 or later.

Complete the following steps to install the self-provisioning plugin on your OpenShift cluster.

1. Ensure that the AMQ Broker Operator and the cert-Manager Operator for OpenShift are installed on the cluster.
2. Log in to your OpenShift cluster by using the OpenShift command-line interface (CLI).
3. In your web browser, navigate to the [Software Downloads page for AMQ Broker 7.14 releases](#).
4. Next to the AMQ Broker 7.14.0 Self-Provisioning Plugin Installation files, click **Download**.
5. Move the downloaded archive to a new directory. The following example moves the archive to a directory called `~/broker/spp`.

```
$ mkdir ~/broker/spp
$ mv amq-broker-self-provisioning-plugin-7.14.0-install-rhel9.zip ~/broker/spp
```

6. Extract the contents of the archives. For example:

```
$ cd ~/broker/spp
$ unzip amq-broker-self-provisioning-plugin-7.14.0-install-rhel9.zip
```

7. Switch to the directory that was created when you extracted the self-provisioning plugin archive.

```
$ cd ../amq-broker-self-provisioning-plugin-7.14.0-install
```

8. Run the script to install the self-provisioning plugin:

```
// AsciiDocDITA.BlockTitle, warning, Block titles can only be assigned to examples, figures,  
and tables in DITA.  
./deploy-plugin.sh
```

For information about how to try out the plugin see the [self-provisioning plugin readme file](#).

CHAPTER 7. FIXED ISSUES

For a complete list of issues that have been fixed in the release, see [AMQ Broker 7.14.0 Fixed Issues](#).

CHAPTER 8. FIXED COMMON VULNERABILITIES AND EXPOSURES

This section details Common Vulnerabilities and Exposures (CVEs) fixed in the AMQ Broker 7.14 release.

- [ENTMQBR-10391](#) - CVE-2026-1605 jetty-server: Eclipse Jetty: Denial of Service due to unreleased JDK Inflator from compressed HTTP requests [amq-7.13]
- [ENTMQBR-10441](#) - CVE-2026-32642 artemis-server: Apache Artemis and Apache ActiveMQ Artemis: Unauthorized address creation due to incorrect authorization during JMS topic subscription. [amq-7.14]
- [ENTMQBR-10443](#) - CVE-2026-24308 zookeeper: Apache ZooKeeper: Information disclosure via improper handling of configuration values [amq-7.14]
- [ENTMQBR-10444](#) - CVE-2026-24281 zookeeper: Apache ZooKeeper: Impersonation of servers or clients via reverse DNS spoofing [amq-7.14]
- [ENTMQBR-10452](#) - CVE-2026-33870 netty-codec-http: Netty: Request smuggling via incorrect parsing of HTTP/1.1 chunked transfer encoding extension values [amq-7.14]
- [ENTMQBR-10458](#) - CVE-2026-33871 netty-codec-http: Netty: Denial of Service via HTTP/2 CONTINUATION frame flood [amq-7.14]
- [ENTMQBR-10463](#) - CVE-2026-33871 netty-codec-http2: Netty: Denial of Service via HTTP/2 CONTINUATION frame flood [amq-7.14]

CHAPTER 9. KNOWN ISSUES

This section describes known issues in AMQ Broker 7.14.

- **ENTMQBR-10446 - Encrypted jdbc connection URL is exposed in logs**
If you encrypt the URL that the broker uses to connect to a JDBC database in the broker configuration file, an unencrypted URL is exposed in the broker logs.
- **ENTMQBR-9810 - Using Java23 results in UnsupportedOperationException: getSubject is supported only if a security manager is allowed**
When you use AMQ Broker 7.14 with Java 23, the following error is displayed:

java.lang.UnsupportedOperationException: getSubject is supported only if a security manager is allowed at java.base/javax.security.auth.Subject.getSubject(Subject.java:347) at

To resolve this issue, add a new argument, **-Djava.security.manager=allow**, to the **JAVA_ARGS** list of Java system arguments in the **artemis.profile** file. This argument enables support for a security manager without actually using one.

- **ENTMQBR-9791 - Failed startup of console when install dir has unicode characters.**
If you rename the default directory, created when you extracted the broker installation archive, to a directory name that contains unicode characters, the AMQ Management Console is unable to start. The following error is displayed in the logging information shown when you start the broker:

**WARN [org.eclipse.jetty.ee9.webapp.WebAppContext] Failed startup of context oeje9w.WebAppContext@626d2016{/metrics,/metrics,null,false,@Connector-0} {/home/jcliffor/Downloads/<directory name>/web/metrics.war}__
java.lang.IllegalArgumentException: Bad escape**

- **ENTMQBR-9109 - Changing ingressDomain doesn't update Routes.**
If you change the value of the **spec.ingressDomain** attribute to an empty value, the domain value is not updated in existing routes. This issue is due to a limitation of the OpenShift routes whereby the **spec.host** field of the OpenShift routes is auto-generated if it is empty only when a new route is created.
- **ENTMQBR-9187 - The Operator shows forbidden errors when you deploy a CR and message migration does not work as expected.**
If you install the Operator by using the OpenShift CLI or using OperatorHub with custom YAML files, you can configure the Operator to watch multiple namespaces or a single namespace that is different from the one in which the Operator is installed. In both of these configurations, the following error is displayed when you deploy the **ActiveMQArtemis** CR:

W0614 14:18:59.355936 1 reflector.go:535] k8s.io/client-go/informers/factory.go:150: failed to list *v1.StatefulSet: statefulSets.apps is forbidden: User "system:serviceaccount:amq-broker-operator-olm:amq-broker-controller-manager" cannot list resource "statefulSets" in API group "apps" at the cluster scope

The broker deployment is created and works as normal. However, if you scale down the deployment subsequently, messages on the broker that was removed are not migrated to another broker and are lost.

To avoid this issue, allow the Operator to watch all namespaces, which is the default, or configure the Operator to watch the single namespace in which the Operator is installed.

- **ENTMQBR-8106 - AMQ Broker Drainer pod doesn't function properly after changing MessageMigration in CR**

You cannot change the value of the **messageMigration** attribute in a running broker deployment. To work around this issue, you must set the required value for the **messageMigration** attribute in a new **ActiveMQ Artemis** CR and create a new broker deployment.

- **ENTMQBR-7359 - Change to current handling of credential secret with 7.10.0 Operator**

The Operator stores the administrator username and password for connecting to the broker in a secret. The default secret name is in the form **<custom-resource-name>credentials-secret**. You can create a secret manually or allow the Operator to create a secret.

If the **adminUser** and **adminPassword** attributes are configured in a Custom Resource before 7.10.0, the Operator updates a manually-created secret with the values of these attributes. Starting in 7.10.0, the Operator no longer updates a secret that was created manually. Therefore, if you change the values of the **adminUser** and **adminPassword** attributes in the CR, you must either: f * Update the secret with the new username and password * Delete the secret and allow the Operator to create a secret. When the Operator creates a secret, it adds the values of the **adminUser** and **adminPassword** attributes if these are specified in the CR. If these attributes are not in the CR, the Operator generates random credentials for the secret.

- **ENTMQBR-7111 - 7.10 versions of operator tend to remove StatefulSet during upgrade**

If you are upgrading to or from AMQ Broker Operator 7.10.0, the new Operator automatically deletes the existing StatefulSet for each deployment during the reconciliation process. When the Operator deletes the StatefulSet, the existing broker pods are deleted, which causes a temporary broker outage.

You can work around this issue by running the following command to manually delete the StatefulSet and orphan the running pods before the Operator gets to delete the StatefulSet:

```
oc delete statefulset <StatefulSet-name> --cascade=orphan
```

Manually deleting the StatefulSet during the upgrade process allows the new Operator to reconcile the StatefulSet without deleting the running pods. For more information, see [Upgrading the Operator using OperatorHub](#) in *Deploying AMQ Broker on OpenShift*.

- **ENTMQBR-5749 - Remove unsupported operators that are visible in OperatorHub**

Only the Operators and Operator channels mentioned in [Deploying the Operator from OperatorHub](#) are supported. For technical reasons associated with Operator publication, other Operator and channels are visible in the OperatorHub and should be ignored. For reference, the following list shows which Operators are visible, but not supported:

- Red Hat Integration - AMQ Broker LTS - all channels
- Red Hat Integration - AMQ Broker - alpha, current, and current-76

- **ENTMQBR-4141 - AMQ Broker Operator: Increasing Persistent Volume size requires manual involvement even after recreating StatefulSet**

If you try to increase the size of the Persistent Volume Claim (PVC) required by brokers in a deployment for persistent storage, the change does not take effect without further manual steps. For example, suppose that you configure the **storage.size** property of a Custom Resource (CR) instance to specify an initial size for the PVC. If you modify the CR to specify a *different* value of **storage.size**, the existing brokers continue to use the original PVC size. This is the case even if you scale the deployment down to zero brokers and then back up to the original number. However, if you scale the size of the deployment up to add additional brokers, the new brokers use the new PVC size.

To work around this issue, and ensure that all brokers in the deployment use the same PVC size, use the OpenShift Container Platform web console to expand the PVC size used by the deployment. In the console, click menu:Storage[Persistent Volume Claims]. Click your deployment. On the **Actions** drop-down menu in the upper-right corner, select **Expand PVC** and enter a new value.

CHAPTER 10. IMPORTANT LINKS

- [Red Hat AMQ Broker 7.13 Release Notes](#)
- [Red Hat AMQ Broker 7.12 Release Notes](#)
- [Red Hat AMQ Broker 7.11 Release Notes](#)
- [Red Hat AMQ Broker 7.10 Release Notes](#)
- [Red Hat AMQ Broker 7.9 Release Notes](#)
- [Red Hat AMQ Broker 7.8 Release Notes](#)
- [Red Hat AMQ Broker 7.7 Release Notes](#)
- [Red Hat AMQ Broker 7.6 Release Notes](#)
- [Red Hat AMQ Broker 7.1 to 7.5 Release Notes \(aggregated\)](#)
- [Red Hat AMQ 7 Supported Configurations](#)
- [Red Hat AMQ 7 Component Details](#)

Revised on 2026-06-03 08:32:40 UTC